

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 7_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Raj is curious about how old he is in the current year.

He has asked you to create a simple program that calculates a person's age based on their birth year. You decide to implement this functionality using the AgeCalculator interface and the HumanAgeCalculator class.

Note: The current year is 2024. Calculate the current age by using the formula: current year - birth year.

Input Format

The input consists of an integer representing the birth year.

Output Format

The output displays "You are X years old." where X is an integer representing the calculated age based on the entered birth year.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1934

Output: You are 90 years old.

Answer

```
import java.util.Scanner;
```

```
// You are using Java
```

```
import java.util.Scanner;
```

```
interface AgeCalculator {  
    int calculateAge(int birthYear);  
}
```

```
class HumanAgeCalculator implements AgeCalculator {  
    private static final int CURRENT_YEAR = 2024;  
  
    @Override  
    public int calculateAge(int birthYear) {  
        return CURRENT_YEAR - birthYear;  
    }  
}
```

```
// Main application class
```

```
class AgeCalculatorApp {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        AgeCalculator ageCalculator = new HumanAgeCalculator();  
  
        int birthYear = scanner.nextInt();
```

```
        int age = ageCalculator.calculateAge(birthYear);  
        System.out.println("You are " + age + " years old.");  
    }  
}
```

Status : Correct

Marks : 10/10